MILITARY SPECIFICATION

AMPLIFIER-CONVERTER: AM-6879/URC

This specification is approved for use by the U. S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, production and test requirements for the Amplifier-Converter AM-6879/URC.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-Q-9858	Quality Program Requirements
MIL-C-12520	Connector, Plug and Receptacle (Electrical, Waterproof) and Accessories, General Specification for
MIL-P-12606	Loudspeaker, Permanent Magnet (Unencased, 3 Inch and 4 Inch Diameter Cone, 2 Watt, Fungus, Gunblast and Immersion Resistant), Type LL-445/U, M12606-01, and M12606-02
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packaging of
MIL-C-39012	Connector, Coaxial, Radio Frequency, General Specification for
MIL-H-49078	Handset H-250 ( )/U
MIL-E-52798	Enamel, Alkyd, Camouflage
MIL-C-55116	Connector, Miniature Audio, Five-pin
MIL-P-55149/9	Post, Binding, Electrical, Waterproof and Nonwaterproof (Uninsulated), Types Pb09nc01 to Pb09nc07, Inclusive

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Headquarters Marine Corps, Code LMA-1, Washington, D.C. 20380 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-463	Definition and System of Units, Electromagnetic Interference Technology
MIL-STD-470	Maintainability Program Requirements (For Systems and Equipments)
MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-1275	Electrical Circuit, 28 Volt DC Transient Characteristics for Military Vehicles

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

HANDBOOKS

Military

MIL-HDBK-472	Maintainability Prediction

DRAWINGS

U.S. Marine Corps

755002A1000	Key Telegraph
755002A1650	Mounting Tray
755002A1810	Cable, CX-13104/URC
755002A1811	Cable, CX-13105/URC
755002A7773 -2	6 pin Audio Connector
755002A7815 -1	Connector to AM6545/GRC-193
755002A7815 -2	Connector to RT-1209/URC

Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

3.1 First article. Prior to submission of any complete production items for acceptance, the contractor shall furnish four first articles for determination of conformance to this specification. Examination and tests shall be those specified herein. Approval of the first articles by the procuring activity shall not relieve the contractor of his obligation to supply equipments conforming to this specification. Any changes or deviations of the production items from the first articles shall be subject to the approval of the contracting officer.

3.2 Material

3.2.1 Materials, processes and parts. Unless otherwise specified (see 6.2.1), all parts, materials, and processes used in construction of this equipment shall conform to MIL-STD-454. The selection of the class, grade, or type shall be such that when mounted in the equipment, the part or material will perform its intended function.

3.2.2 Integrated electronics, packaging. All solid state devices and all microcircuits employed shall be packaged in hermetically sealed packages. No plastic (for example, epoxy, silicon, organo-metallics, phenolics, and so forth) encapsulated devices shall be accepted without specific written approval of the procuring activity.

3.3 Design and construction.

3.3.1 Major components. The following components and ancillary items are part of, or used with, the Amplifier-Converter, AM-6879/URC and shall be supplied by the contractor:

Item	Quantity
Amplifier-Converter	1 each
Handset, H-250/U	1 each
Telegraph Key KY-872/PRC-104	1 each
Cable Assembly. Special Purpose, Electrical CX-13104/URC	1 each
Cable Assembly, Special Purpose, Electrical CX-13105/URC	1 each
Mounting Tray, Electrical Equipment MT-4874/URC	1 each

3.3.1.1 Amplifier-converter. The amplifier-converter shall interface with the receiver-transmitter RT-1209/URC, the radio frequency amplifier, AM-6545/GRC-193, and teletype equipment such as the AN/TGC-14(V) as installed in the AN/GRC-193 transportable base station, high frequency (HF), single sideband (SSB) radio system. The amplifier-converter provides loudspeaker output with optional squelch, conversion from 20 milliampere (mA) or 60 mA teletypewriter (TTY) direct current (DC) loop to wide deviation frequency shift-keying (FSK) (and vice-versa) and power line and control line electromagnetic interference (EMI) filtering and conditioning to protect the RT-1209/URC from transients and radio frequency interference (RFI).

3.3.1.1.1 Control elements. The following controls shall be provided and located on the front panel.

3.3.1.1.1.1 TTY/loudspeaker switch. A single switch shall be provided to turn on or off either the TTY circuits or speaker (audio and squelch circuit).

3.3.1.1.1.2 Squelch. A control shall be provided to both turn the squelch circuit on or off and to adjust the squelch threshold.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

3.3.1.1.1.3 Loop current adjust. A locking type screwdriver control shall be provided for adjustment of the DC loop current.

3.3.1.1.1.4 Loop current select. A switch shall be provided to select either 60 mA or 20 mA DC loop operation.

3.3.1.1.2 Front panel connectors. The following connectors shall be provided on the front panel.

3.3.1.1.2.1 Binding posts. Four binding posts, type PB09NC04, in accordance with MIL-P-55149/9, shall be provided for TTY key and DC loop interface to the TTY.

3.3.1.1.2.2 Handset. A 5 pin audio connector, type UG-183/U, in accordance with MIL-C-55116, shall be provided for connection of the H-250/U handset.

3.3.1.1.2.3 RT-1209/URC front. A 6 pin audio connector, in accordance with Drawing No. 755002A7773-2, shall be provided for audio and keying interface with receiver-transmitter RT-1209/URC or the secure voice equipment KY-65.

3.3.1.1.3 Fuseholders. Two fuseholders shall be provided. The active fuseholder shall be located in the front panel of the unit and the spare fuseholder on the rear surface of the unit.

3.3.1.1.4 Test points TTY. Test points for measurement of the loop current shall be provided on the front panel.

3.3.1.1.5 Loudspeaker. A loudspeaker shall be provided on the front panel of the unit. It shall be front facing, 2-watt (W), permanent magnet, resistant to fungus, gunblast and immersion and equivalent to type LS-445/U of MIL-L-12606.

3.3.1.1.6 Rear mounted connectors. The following connectors shall be mounted on the rear surface.

3.3.1.1.6.1 RT-1209/URC rear. A male receptacle in accordance with Drawing No. 755002A7815-2 and MIL-C-12520 shall be provided for interface to the RT-1209/URC. This interface shall be as defined in Table I.

3.3.1.1.6.2 AM-6545/GRC-193 control and power. A male receptacle in accordance with Drawing No. 755002A7815-1 and MIL-C-12520 shall be provided for interface to the AM-6545/GRC-193. The interface shall be as defined in Table II.

3.3.1.1.6.3 AM-6545/GRC-193 Radio Frequency (RF). A female type N receptacle in accordance with MIL-C-39012 shall provide RF input/output interface to the AM-6545/GRC-193.

3.3.1.1.7 Reliability. The amplifier-converter shall possess a specified mean-time between-failure (MTBF), θ_0, of no less than 4400 hours, and a minimum acceptable MTBF of no less than 2200 hours over the specified environmental operating range.

3.3.1.1.8 Reverse polarity. The amplifier-converter shall incorporate reverse polarity protection by the use of protective diodes.

3.3.1.1.9 Ground. The chassis shall be negative.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

Table I. RT-1209/URC Interface definition

INTERFACE CONNECTOR PIN NO.	LINE FUNCTION	LINE SOURCE	LINE CONDITION V.S. STATUS	OPERATING LIMITS
38, 39	+26.5 VDC (Nom.)	PA	Direct connection to power source	2 pins RX - 200 mA nom. TX - 0.6 A max. 20 - 32 VDC
25, 26	GND (DC Return)	PA	Direct connection to chassis ground	2 pins Chassis is negative
32	Shield Ground (Cable Shield)	NA	Direct connection to chassis ground	
14 20 35 29 12 13	Freq. Code, 10 MHz (2) 10 MHz (1) 1 MHz (8) 1 MHz (4) 1 MHz (2) 1 MHz (1)	RT	0 logic - Ground through diode and switch. 200 ohms maximum 1 logic - Open (true) 50,000 ohms minimum to ground	0 = 5 mA max. Max. pullup voltage = 5.25 VDC
30	PA ON/OFF	RT	0 - Ground through diode and CE junction of NPN transistor OFF - Open when RT is in RX only mode or OFF	100 mA max. 32V max.
45	System ON/OFF	RT	ON - +26.5 VDC when RT is "ON" in any mode OFF - 0 VDC, direct connection to RT power source input circuits	0.3 A max. No voltage from PA allowed

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

Table I. RT-1209/URC Interface definition - Continued

INTERFACE CONNECTOR PIN NO.	LINE FUNCTION	LINE SOURCE	LINE CONDITION V.S. STATUS	OPERATING LIMITS
4	PA Keyline	RT	Keyed - Ground through CE junction of NPN transistor Not keyed - Open	50 mA max. 32 V max. 50K ohms min.
10	Tune in Progress	PA	Ground during automatic tune cycle +6V during normal RX/TX	1 mA max. 2.5K ohms max.
33	Tune Start	RT	Ground pulse (30 ms. min.) through CE junction of NPN transistor when: (a) any frequency digit (except 100 Hz) is changed; (b) RT is switched "ON" or (c) RT mode selector is switched from RX only to TR position (but NOT in reverse).	Pullup voltage - 26VDC (nom.) through 12K ohms Note: Each user of this line shall use positive logic and diode isolation to present an open circuit to ground in the absence of tune start
44	Tune Fault	PA	Not Fault - Ground Fault - Open	2.5K ohms max. 6 VDC pullup 2 mA max.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

Table I. RT-1209/URC Interface definition - Continued

INTERFACE CONNECTOR PIN NO.	LINE FUNCTION	LINE SOURCE	LINE CONDITION V.S. STATUS	OPERATING LIMITS
8	ALC (Xmit Gain Control)	PA	Positive going analog of RF output power above adjustable threshold. DC source impedance must be less than 2K ohms	Zero + 0.6VDC offset provides no RF attenuation +6 VDC provides NLT 37 dB attenuation. Attenuation curve is monotonic with linearity of 8 + dB per V
40	Sidetone enable	PA	Sidetone - Ground Not sidetone - Open	2.5K ohms max. 6 VDC pullup 1 mA max. Note: This line must remain open in receive only (PA OFF) modes.
5	V forward	RT	Positive going voltage proportional to forward power at output of harmonic filter.	10K source impedance.
41	V reflected	RT	Negative going voltage proportional to reflected power at output of harmonic filter.	10K source impedance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

Table I. RT-1209/URC Interface definition - Continued

INTERFACE CONNECTOR PIN NO.	LINE FUNCTION	LINE SOURCE	LINE CONDITION V.S. STATUS	OPERATING LIMITS
19	Keyline	RT	Keyed - Ground through diode and CE junction of NPN transistor Not keyed - Open	50 mA max. 32V max. 50K ohms min.
2	RF IN/OUT	NA	RF signal to and from mod. /demod. module of RT	50 Ohms nominal
1,3,6,7	Signal return (gnd)	NA	Shielding pins for RF IN/ OUT above (pin 2)	
16	RF IN	NA	RF signal into harmonic filter module in Xmit mode	Jumpered to pin 2 in interfacing connector when PA power output is greater than 20W. 50 ohms nominal
9,15,21, 22	Signal return (gnd)	NA	Shielding pins for RF IN above (pin 16)	
47	RF OUT [RF drive to PA (0.25W) or Ant. coupler (20W)]	NA	RF signal out of harmonic filter module in Xmit mode.	50 ohms nominal
42,43,46, 48	Signal return (gnd)	NA	Shielding pins for RF OUT above (pin 47)	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

Table I. RT-1209/URC Interface definition - Continued

INTERFACE CONNECTOR PIN NO.	LINE FUNCTION	LINE SOURCE	LINE CONDITION V.S. STATUS	OPERATING LIMITS
23	EXT. FREQ. STD. (5 MHz RF input)	PA	This input normally shorted to ground. To utilize external frequency standard remove jumpers E1 to E2, E3 to E4, and E5 to E6 in synthesizer module. Install jumper E4 to E5.	0 + 0.5 dBm 50 ohms nominal.
36	+6.5 VDC	RT	+6.5V when pin 45 is high (+28V ON)	350 mA max. This line may be used only by the AM-6874/ PRC-104
37	+12.5 VDC	RT	+12.5V when pin 45 is high (+28V ON)	50 mA max. This line may be used Only by the AM-6874/ PRC-104.
11	2 - 3 MHz Band	RT	GND through CE junction of NPN transistor when selected frequency is in 2.0000 to 2.9999 MHz range Otherwise: +28V through 2970 ohm relay.	50 mA max. Diode isolation required.
31,34	SPARE	NA		
17,18,24 27,28	Signal return (gnd)	NA	Shielding pins for Pin 23	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

Table II. AM-6545/GRC-193 Interface definition

INTERFACE CONNECTOR PIN NO.	LINE FUNCTION	LINE SOURCE	LINE CONDITION V.S. STATUS	OPERATING LIMITS
L M N P R S	Freq. Code, 1 MHz (8) Freq. Code, 1 MHz (4) Freq. Code, 1 MHz (2) Freq. Code, 1 MHz (1) Freq. Code, 10 MHz (2) Freq. Code, 10 MHz (1)	R/T R/T R/T R/T R/T	0 Logic - Ground through diode and switch or C-E junction of NPN transistor 1 Logic - Open (true)	5 mA max. 200 ohms max. 20,000 ohms min. max. pullup = 5.25 VDC
E	Sidetone	PA	When Tx RF output is present-GND through C-E junction of NPN transistor Rx and Tx (no signal) - open	33V max. 5 mA max. 50K ohms min.
K	Tune in Process	PA	During automatic tune cycle GND through relay contacts Normal Rx-Tx operation - open	32V max. 2A max. 50K ohms min.
F	Keyline	R/T	Keyed - GND through C-E junction of NPN transistor Keyed - Open	32V max. 10 mA max. 5K ohms max. 200K ohms min.
U	Primary Power (+26.5 VDC Nominal)	PA	Direct connection to positive side of power source	per MIL-STD-1275A (AT)
A	GND	NA	Direct connection to chassis and negative side of power source	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

Table II. AM-6545/GRC-193 Interface definition - Continued

INTERFACE CONNECTOR PIN NO.	LINE FUNCTION	LINE SOURCE	LINE CONDITION V.S. STATUS	OPERATING LIMITS
T	PA ON/OFF	R/T	R/T ON and TR mode selected GND through diode and C-E junction of NPN transistor R/T OFF or mode selector in either Rx only position - open	32V max. 8 mA max. 200K ohms min.
D	Power ON	R/T	R/T ON (all modes) - 26.5V nominal. R/T OFF - 0 VDC	+20 to 32 VDC 10 mA max. No voltage from PA allowed
C	Tune Start	R/T	GND pulse (30 mS min.)through C-E junction of NPN transistor when: (a) Any frequency digit (except 100 Hz) is changed (b) R/T is switched "ON"; or (c) R/T mode selector is switched from "Rx only" to "TR" position (but NOT in reverse)	26 VDC pullup through 10K ohms. 13 mA max. NOTE: Each user of this line shall use positive logic and diode isolation to present an open circuit to ground in the absence of tune start
J	ALC (Xmit gain control)	PA	Positive going analog of RF output power above adjustable threshold. DC source impedance less than 2K ohms	0 to 0.6 VDC provides no RF attentuation. +6 VDC provides NLT 40 dB RF drive reduction. Attenuation curve is monotonic with linearity of 8 + 2 dB per volt

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

Table II. AM-6545/GRC-193 Interface definition - Continued

INTERFACE CONNECTOR PIN NO.	LINE FUNCTION	LINE SOURCE	LINE CONDITION V.S. STATUS	OPERATING LIMITS
H	Fault	PA	Fault - GND (2500 ohms max.) No Fault - OPEN	6 VDC pullup 2 mA max. 200K ohms min.
B	Shield GND	NA	RF ground through 0.1 uF capacitor	
V, W	Spare	NA		
Type N Coaxial	RF-IN/OUT	NA	Rx - Rcvr input Tx - RF drive to PA	50 ohms nominal Nominal 0.25W - 2 dB, +4 dB and subject to reduction by ALC action

Source: https://assist.dla.mil - Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

3.3.1.1.10 Test points. The equipment shall incorporate test points to facilitate fault isolation and repair.

3.3.1.1.11 Compatibility. The equipment shall be compatible with Handset H-250/U, and the Remote Control Group AN/GRA-39B.

3.3.1.1.12 Interchangeability. Provisions for interchangeability are required and shall be in accordance with requirement 7 of MIL-STD-454.

3.3.1.1.13 Warm-up time. The amplifier-converter shall require no warm-up time.

3.3.1.1.14 Duty cycle. The amplifier-converter shall be capable of continuous use.

3.3.1.2 Handset H-250/U. The handset shall be in accordance with MIL-H-49078.

3.3.1.3 Telegraph Key KY-872/PRC-104. The telegraph key shall be in accordance with Drawing No. 755002A1000.

3.3.1.4 Vehicle mount to receiver-transmitter (RT) cable CX-13104/URC. This cable shall interconnect the amplifier-converter to the RT-1209/URC and shall be in accordance with Drawing No. 755002A1810.

3.3.1.5 Audio and Keying cable CX-13105/URC. This cable shall be provide the audio and keying interface between the amplifier-converter and the RT-1209/URC and shall be in accordance with Drawing No. 755002A1811.

3.3.1.6 Mounting tray MT-4874/URC. The mounting tray shall support the AM-6879/URC and RT-1209/URC in the radio set AN/GRC-193 configuration. Mount shall be in accordance with Drawing No. 755002A1650.

3.3.2 Control life. All controls shall be functioned end-to-end without failure for no less than 10,000 cycles.

3.3.3 Control end stops. All controls identified as having end stops shall be capable of withstanding repeat force in each direction of 48 inch-ounces without permanent deformation to end stops. Frequency selectors shall not be actuated with less than 2 pounds of force, nor require more than 6 pounds for activation. Push type dial lamp switch shall not be activated with less than 1 pound of force nor require more than 6 pounds for activation.

3.3.4 Covers and seals. Amplifier-converter covers, seals or sealing surfaces shall be capable of 25 open and close cycles without loss of leakage resistance.

3.3.5 Soldering. Soldering shall be in accordance with Requirement 5 of MIL-STD-454.

3.3.6 Mechanical characteristics. The amplifier-converter shall be essentially a rectangular box fabricated from aluminum for reasonable weight. The unit shall be watertight and ruggedized for rough handling and hard mounting in vehicle installations. Mounting flanges and bosses on the bottom surface shall provide for retention to a vehicle mounting tray alongside the receiver-exciter. Provisions are also required for adapter plate mounting of the receiver-exciter

Source: https://assist.dla.mil - Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

unit atop the amplifier-converter for future installations that will add a preselector/remote control unit alongside on the same mounting tray. All controls, loudspeaker, TTY terminals, fuse, and audio connectors shall be front panel mounted. Interface connectors shall be rear panel mounted. The amplifier-converter to receiver-exciter audio and interface jumper cables shall be retained. Projecting flanges shall protect the front panel controls and connectors. The amplifier-converter shall be constructed as a two-piece enclosure consisting of a rear cover and a panel/enclosure assembly. All electronic circuitry, except front panel controls shall be carried on plug-in modules that are hardware-mounted to the enclosure. Cover screws shall be retained.

3.4 Maintainability. Design criteria for the maintainability of the equipment components, and ancillary items to be contractor-furnished shall be in accordance with MIL-STD-470, MIL-STD-471, and MIL-HDBK-472. The mean time to repair (MTTR) for the replacement of defective modules shall be no more than 15 minutes. The MTTR for the amplifier-converter, modules and subassemblies, other than throwaway items, at the depot levels, shall be no more than 3 hours. Comformance to the specified MTTR shall be certified by a demonstration test using personnel with equivalent depot level training.

3.5 Transportability. The amplifier-converter shall be capable of being transported by wheeled or tracked vehicles operating in a tactical environment.

3.6 Performance.

3.6.1 Electrical characteristics.

3.6.1.1 Input power. Primary input power shall be 28 direct current volts (VDC) nominal conforming to MIL-STD-1275 and shall operate over the range of 22.0 to 30 VDC.

3.6.1.2 Transient protection. All voltage transients allowed by MIL-STD-1275 shall be processed such that input power applied to the amplifier-converter and through it to the RT-1209/URC shall not exceed 36 volts (V) for more than 100 microseconds per transient.

3.6.1.3 Loudspeaker amplifier.

3.6.1.3.1 Loudspeaker. The loudspeaker shall be front facing, 2 W, permanent magnet, resistant to fungus, gunblast and immersion, and equivalent to type LS-445/U of MIL-L-12606.

3.6.1.3.2 Amplifier.

3.6.1.3.2.1 Amplifier input. Audio frequency (AF) input, from the RT-1209/URC, shall be a minimum of 2 milliwatt (mW) into 500 ohms. Distortion of the RT output is 5 percent maximum. The loudspeaker volume is controlled by the RT volume control.

3.6.1.3.2.2 Amplifier output. Minimum AF output over the 300 to 3000 hertz (Hz) range shall be 1 W into an 8 ohm loudspeaker. Total harmonic distortion of the RT output is 5 percent maximum. The loudspeaker volume is controlled by the RT volume control.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

3.6.1.4 Squelch. Squelched/unsquelched AF output modes shall be applicable to both the loudspeaker and the handset connector. The squelch system shall provide maximum discrimination against unsquelching on nonvoice signals such as noise, radio teletypewriter (RTTY), or telegraphy by recognizing the syllabic low frequency content of voice signals.

3.6.1.4.1 AF reduction. In squelched condition, the AF output reduction shall be a minimum of 30 decibels (dB).

3.6.1.4.2 Squelch sensitivity. The AF output shall unsquelch with voice signals having a minimum SINAD of 10 dB. A squelch sensitivity control shall be provided to allow the squelch/unsquelch threshold to be increased to a 20 dB SINAD. A squelch on/off switch shall also be provided.

3.6.1.4.3 Squelch hysteresis. After unsquelching, the unsquelched condition shall be retained for a SINAD reduction up to 2 dB.

3.6.1.4.4 Attack and release times. The unsquelch attack time shall be a maximum of 35 milliseconds (ms). The release time to the squelched condition shall be delayed at least 300 ms after termination of the voice signal.

3.6.1.5 Teletype converter. The unit shall incorporate a modular, plug-in, single channel, half-duplex FSK to DC loop converter. Deletion of this module shall not affect any other function of the unit. The TTY on/off switch shall control the power to the converter circuitry and, in the OFF position, disconnect the converter from the audio input/output of the receiver-transmitter.

3.6.1.5.1 Receiver input/output. In TTY receive mode, the converter shall accept a 10 dB SINAD FSK signal at 0 dB referred to 1 mW (dBm) level across 600 ohms as obtained from the RT-1209/URC in the data mode. The converter shall produce a mark output of 20 mA (or 60 mA) into the DC loop when a 1575 ±15 Hz signal is present and open circuit the DC loop (less than 2 mA) when a 2425 ±15 Hz signal is present. The DC loop signal shall be free of blocking or spurious transients up to a rate of 75 baud.

3.6.1.5.2 DC loop characteristics. Circuitry and controls shall be provided to permit TTY DC loop operation at either 20 ±4 or 60 ±12 mA and shall be externally adjustable. Test points for measurement of the loop current shall be provided.

3.6.1.5.3 Transmit input/output. In TTY transmit mode, obtained by switch closure across the transmit/receive binding posts, the converter shall output a 1575 ±5 Hz signal when the DC loop is closed (20 or 60 mA) and an output of 2425 ± 5 Hz when the DC loop is open. Output level shall be 0 dBm ±4 dB into a 600-ohm load. Output signal shall be free of blocking or spurious transients at transmission rates up to 75 baud. Frequency accuracy of tones shall be ±15 Hz at temperature extremes.

3.6.1.5.4 Remoted teletypewriter operation. The transmit/receive switching and half-duplex DC loop data transfer shall be capable of operation when the TTY, AN/TGC-14A(V) is remoted with two pairs of field wire, WD-1/TT, for distances up to 1 mile.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

3.6.2 Burn-in. The amplifier-converter shall be able to withstand the burn-in test as specified in 4.8.32 prior to acceptance for delivery.

3.6.3 Random vibration test. The amplifier-converter shall be able to withstand the vibration test as specified in 4.8.33 prior to acceptance for delivery.

3.6.4 Temperature cycling. The amplifier-converter shall be able to withstand the temperature cycling test as specified in 4.8.34 prior to acceptance for delivery.

3.7 Environmental requirements.

3.7.1 Service conditions. The amplifier-converter shall be capable of meeting the requirements specified herein under the following service conditions.

(a)	Low temperature operating	-40°C (-40°F)
(b)	High temperature operating	+55°C (+131°F)
(c)	Humidity	up to 98 percent
(d)	Altitude	Sea Level to 3048 meters (10,000 feet)
(e)	Vibration	
	1 Vibration, Secured Cargo	Transported as secured cargo by rail, sea, truck or semitrailer.
	2 Vibration installed	Installed in ground vehicles, wheeled (utility vehicle M151A2), 6000 miles rated mileage. Frequency range limited to 55 Hz.
(f)	Shock	
	1 (Bench handling)	The equipment shall be capable of meeting performance requirements after the normal handling required for bench testing.
	2 (Basic Design)	The equipment shall meet the performance requirements with no more than superficial damage after exposure to shocks of 15 gravities, 11 MS duration, 1/2 sine wave form.
	3 (Transit)	The equipment as packaged for shipment shall meet the performance requirements with no more than superficial damage after a series of drops from a height of 1.2192 meters (4 ft) to a surface of 5 centimeters (2 in) of plywood backed by concrete.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

(g)	Dust	Dry dust (fine sand) laden atmosphere at air velocities up to 609.6 meters (2000 ft) per minute.
(h)	Leakage	Submergence to a depth of 0.9144 meters (3 ft) of water without leakage.
(i)	Rain	Blowing rain at rates up to 0.127 meters (5 in) per hour at wind velocities up to 64.37 meters per hour (40 miles per hour (MPH)).
(j)	Salt Fog	The equipment shall resist the effect of a salt laden atmosphere typical of coastal areas.
(k)	Fungus	The equipment shall not support the growth of fungi under conditions of high humidity, warm atmosphere and the presence of inorganic salts.

3.7.2 Storage conditions. The equipment shall be capable of being stored under the following conditions without degradation.

(a) Ambient temperature range. From -62°C (-80°F) to +71°C (+160°F)

(b) Altitude. From Sea level to 50,000 feet

3.7.3 Orientation. The equipment shall be capable of operation at any orientation.

3.8 Electromagnetic compatibility (EMC). The equipment shall meet the interference control limits of MIL-STD-461, using the test methods of MIL-STD-462. Terms and conditions shall be in accordance with MIL-STD-463. Filtering of the interface leads (power, audio, and control) listed in Table II and at the audio connectors shall be sufficient to avoid self-induced radio frequency interference in a 400-W system installation in an M151A2 utility vehicle with a 16 or 32 foot whip antenna. This unit shall also provide the lead filtering for the RT-1209/URC in the system.

3.9 Dimensions. The dimensions of the amplifier-converter shall be 10.5 inches (26.7 centimeters (cm)) wide, 6.6 inches (16.8 cm) high and 6.06 inches (15.4 cm) in depth.

3.10 Weight. The weight of the amplifier-converter, including the retained cables, shall not exceed 16 pounds (7.26 kilograms (Kg)).

3.11 Finish. The radio set shall be finished in accordance with the requirements of MIL-E-52798 including required metal preparation and primers. Exterior color shall be forest green.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

3.12 Marking. Marking shall conform to Requirement 67 of MIL-STD-454.

3.13 Workmanship. All workmanship shall be in accordance with the requirements established in MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the constractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality program requirement. The contractor shall provide and maintain an effective quality program acceptable to the Government for supplies covered herein. The quality program shall be in accordance with MIL-Q-9858.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will include (a) surveillance of the operations to determine that practices, methods, and procedures of the written inspection system are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him, or of which he is notified, shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections and tests. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4)
(b) Production inspection (see 4.5)
(c) Sample inspection (see 4.6)
(d) Environmental inspection (see 4.7)

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all examinations and tests shall be performed under the following conditions:

(a)	Temperature	room ambient
(b)	Altitude	normal ground
(c)	Vibration	none

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

(d)	Humidity	relative humidity up to 90 percent
(e)	Input voltage	28 VDC nominal
(f)	Tolerances	Tolerances except for performance limits are ± 10 percent

4.4 First article inspection. The Government representative present will select which item is to be subjected to each test. The requirements of 4.5.1 and 4.5.2 apply as a prerequisite to performing first article inspection tests. First article inspection shall consist of tests indicated in Table III.

4.5 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the tests and examinations indicated in Table III.

4.5.1 Inspection of parts, subassemblies, and items. Inspections shall be performed to determine conformance to the specified requirements prior to inspection of preparation for delivery. All silicon monolithic and hybrid microcircuits including both digital and linear circuit types shall be subjected to and pass as a minimum the quality assurance tests in accordance with MIL-STD-883, Method 5004, Class B.

TABLE III. Tests

	Test	Requirement Paragraph	Test Paragraph	First Article	Production	Sample	Environmental
1.	Mechanical Inspection	3.3.6	4.5.2	X		X	
2.	Amplifier-Converter	3.3.1.1	4.8.2	X		X	
3.	Control Elements	3.3.1.1.1	4.8.3	X	X		
4.	Front Panel Connectors	3.3.1.1.2	4.8.4	X	X		
5.	Fuse holders	3.3.1.1.3	4.8.5	X	X		
6.	Test Points TTY	3.3.1.1.4	4.8.6	X	X		
7.	Loudspeaker	3.3.1.1 5	4.8.7	X	X		
8.	Rear-mounted Connectors	3.3.1.1.6	4.8.8	X	X		
9.	Reliability	3.3.1.1.7	4.8.9	X			
10.	Reverse Polarity	3.3.1.1.8	4.8.10	X		X	
11.	Ground	3.3.1.1.9	4.8.11	X			
12.	Test Points	3.3.1.1.10	4.8.12	X			
13.	Compatibility	3.3.1.1.11	4.8.13	X		X	
14.	Interchangeability	3.3.1.1.12	4.8.14	X		X	
15.	Warm-up Time	3.3.1.1.13	4.8.15	X		X	
16.	Duty Cycle	3.3.1.1.14	4.8.16	X		X	
17.	Control Life	3.3.2	4.8.17	X			

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

TABLE III. Tests (cont'd)

	Test	Requirement Paragraph	Test Paragraph	First Article	Production Sample	Environmental
18.	Control End Stops	3.3.3	4.8.18	X		
19.	Covers and Seals	3.3.4	4.8.19	X		X
20.	Maintainability	3.4	4.8.20	X		
21.	Input Power	3.6.1.1	4.8.21	X	X	
22.	Transient Protection	3.6.1.2	4.8.22	X	X	
23.	Amplifier Input	3.6.1.3.2.1	4.8.23	X	X	
24.	Amplifier Output	3.6.1.3.2.2	4.8.23	X	X	
25.	Squelch	3.6.1.4	4.8.24	X	X	
26.	AF Reduction	3.6.1.4.1	4.8.24	X	X	
27.	Squelch Sensitivity	3.6.1.4.2	4.4.24	X	X	
28.	Squelch Hysteresis	3.6.1.4.3	4.8.24	X	X	
29.	Attack and Release Times	3.6.1.4.4	4.8.24	X	X	
30.	Receive Input/Output	3.6.1.5.1	4.8.25	X	X	
31.	DC Loop Characteristics	3.6.1.5.2	4.8.26	X	X	
32.	Transmit Input/Output	3.6.1.5.3	4.8.27	X	X	
33.	Remoted Teletype Operation	3.6.1.5.4	4.8.28	X	X	
34.	EMC	3.8	4.8.29	X		
35.	Dimensions	3.9	4.8.30	X	X	
36.	Weight	3.10	4.8.31	X	X	
37.	Burn-in	3.6.2	4.8.32	X	X	
38.	Random Vibration Test	3.6.3	4.8.33		X	
39.	Temperature Cycling	3.6.4	4.8.34		X	
40.	Low Temperature	3.7.1(a)	4.8.1.1	X		X
41.	High Temperture	3.7.1(b)	4.8.1.1	X		X
42.	Humidity	3.7.1(c)	4.8.1.2	X		X
43.	Altitude	3.7.1(d)	4.8.1.3	X		X
44.	Vibration	3.7.1(e)	4.8.1.4.1	X		X
			4.8.1.4.2	X		X
45.	Shock	3.7.1(f)	4.8.1.5.1	X		X
			4.8.1.5.2	X		X
			4.8.1.5.3	X		X
46.	Dust	3.7.1(g)	4.8.1.6	X		X
47.	Leakage (Immersion)	3.7.1(h)	4.8.1.7	X		X
48.	Rain	3.7.1(i)	4.8.1.8	X		X
49.	Salt Fog	3.7.1(j)	4.8.1.9	X		X
50.	Fungus	3.7.1(k)	4.8.1.10	X		X

Note: Sequence of service condition tests shall be in accordance with Table I of MIL-STD-810 for Group E, Type j.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

4.5.2 Mechanical inspection. Each equipment shall be given a thorough mechanical and visual inspection to determine that the quality of all materials and workmanship is in compliance with the requirements specified herein. Particular attention shall be given to the following:

(a) Completeness
(b) Nameplates, identification markings and labels
(c) Ease of operation of connector mating, thumb screws, controls and switches and any other part intended for adjustment, movement or operation
(d) Finishes
(e) Welded joints
(f) Check of solder joints
(g) The fit of components in their respective positions
(h) Check of mounting means
(i) Check of safety features
(j) Fastening and securing devices or parts
(k) Accessibility of components and parts for servicing
(l) Cable run between components including plugs and receptacles
(m) Fit of sealing gasket on cover and wear on gasket
(n) Check of all connector fittings
(o) Test points
(p) Overall dimensions
(q) Weight
(r) Other visual defects

4.6 Sample inspection. The tests in Table III indicated as sample tests shall be performed on samples from production in accordance with the procedures for special inspection levels per MIL-STD-105. Acceptable Quality Level (AQL) for each sample test shall be 6.5. The inspection level shall be S-3 for normal, tightened and reduced inspection.

4.6.1 Noncompliance. If a sample unit fails a test indicated in Table III, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and the details of the corrective action taken to correct units of the same conditions. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. Approval to ship the lot may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. The Government reserves the right to require selection of new samples and a new sample test to be performed after completion of any reworking of the lot.

4.6.2 Corrective action. Corrective action found necessary by the contractor or procuring agency concerned as a result of test failure shall be at the contractor's expense. Further, the contractor shall retrofit the corrective action into the production from which the failed item was originally selected. Applicability of the corrective action and retrofit to previous production shall be determined by the procuring agency. Cost of all retrofits and subsequent reinspection shall be borne by the contractor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

4.7 Environmental inspection. Testing indicated for environmental inspection in Table III shall be performed upon samples drawn from production as follows:

Type I - The following tests shall be performed one time upon samples drawn from the first 100 equipments produced: rain, salt fog, dust and fungus. One sample shall be drawn for each test.

Type II- The following tests shall be performed quarterly upon samples drawn from production as follows:
Sample 1 - Temperature (high and low), humidity and leakage (immersion), and altitude.
Sample 2 - Vibration and shock.

4.7.1 Approval for shipment. Delivery of first production equipments shall be withheld until successful completion of the first type I and type II tests. Upon successful completion, authorization to ship will be granted for a period of 120 days or successful completion of the next quarterly test, whichever occurs first. Continued authorization to ship shall be granted if the quarterly tests are successfully completed. In the event a sample fails, the provisions of 4.6.1 and 4.6.2 shall apply.

4.7.2 Reinspection of conforming environmental sample units. Unless otherwise specified in the contract (see 6.2.1), sample units which have been subjected to, and passed, environmental tests may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all damage.

4.8 Tests.

4.8.1 Environmental.

4.8.1.1 Temperature. Temperature tests shall be in accordance with Procedure I, Method 502.1, and Procedure II, Method 501.1, of MIL-STD-810 and the following:

(a) The low nonoperating (storage) temperature shall be -62°C (-80°F) for 48 hours.
(b) The low-operating temperature shall be -40°C (-40°F).
(c) Where performance measurements are specified, the tests in 4.9 shall be performed.
(d) The high nonoperating and operating temperature shall be 71°C (+160°F) and +55°C (131°F) respectfully.
(e) The duty cycle for the equipment during operating periods shall be 1 minute transmit to 9 minutes receive. All measurements shall be made under these conditions except where a test may be required to start from a power off condition.
(f) Unless otherwise specified, the mode of operation shall be TTY send.

Temperature test procedures shall be as follows:

(a) Make performance measurements at Step 1 of Procedure I, Method 502.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

(b) At Step 5 of Procedure I, Method 502.1, perform the warmup test specified in 4.8.15 at the initial application of input power at 20 VDC input. Make performance measurements.

(c) Testing shall be continuous during transition from low temperature, operating, to high temperature, operating. Starting at a temperature of -40°C (-40°F) the temperature shall be first increased to -20°C (-4°F) and then by 20°F degree steps up to the upper nonoperating limit of 71°C (160°F). The equipment shall be held at each step long enough for temperature stabilization to occur. Performance measurements are made at each operating temperature step with alternate 20 and 32 VDC input voltages except the measurements at -40.0°C (-40°F) are made with a 24 VDC input voltage.

(d) At Step 8, Procedure II, Method 501.1, continue operating the equipment for 4 hours. Make performance measurements with 20 and 32 VDC input voltage.

(e) Make measurements at Step 10, Procedure II, Method 501.1.

4.8.1.2 Humidity. The equipment shall be subjected to the humidity test in accordance with Method 507.1, Procedure III of MIL-STD-810. The equipment shall be operated in all modes with simulated battery where measurements are required.

(a) Make performance measurements at Step 4, Step 6, Step 8 (every 24 hours), and Step 11.

(b) Step 6 performance measurements shall be made during the last 5 hours of the fifth cycle.

(c) For environmental inspection in 4.7, humidity test shall be conducted in accordance with MIL-STD-810, Method 507, Procedure III except that steps 7 and 8 shall be omitted. Performance measurements shall be made during the last 5 hours of the fifth cycle.

4.8.1.3 Altitude. The altitude test shall be performed with the equipment powered by a simulated battery in accordance with procedure I, Method 500.1 of MIL-STD-810.

4.8.1.4 Vibration. The vibration requirements of the amplifier-converter equipment shall be verified by the following tests.

4.8.1.4.1 Vibration, transport. The amplifier-converter shall be rigidly clamped to a fixture and vibration tested in accordance with MIL-STD-810, Method 514.2, Procedure X as applicable to:

Category g - Equipment transported as cargo
Transport Mode - Rail, Air, Sea, Truck or Semi-trailer
Curve Selection - Curve AW , Figure 514.2-7 and resonance below 5 is not expected.
Cycling Time - 84 minutes axis.
Sweep time - 12 minutes sweep from 5 to 200 to 5 Hz.
Resonance Dwell - None required.

The equipment shall be nonoperating during the vibration exposure and after the vibration shall be tested to demonstrate compliance with 4.2. The equipment shall operate within the specified tolerance limits and shall not exhibit damage after the vibration test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

4.8.1.4.2 Vibration, vehicular configuration. The amplifier-converter shall be hard mounted by normal means to a vibration exciter and tested, in accordance with MIL-STD-810, Method 514.2, Procedure VIII as applicable to:

Category f - equipment installed in ground vehicles (Table 514.2-VI).
Equipment Conditions - Wheeled Vehicles.

Curve Selection	Curve Y
Cycling Time/Axis	3 hours
Frequency Range	5 to 55 Hz
Sweep Rate	1 octave/minute nominal

During the vibration test, the following parameters shall be within the specified limits:

Amplifier Output
Squelch

4.8.1.5 Shock.

4.8.1.5.1 Shock (bench handling). The test for bench handling shall be in accordance with Method 516.2, Procedure V, of MIL-STD-810.

4.8.1.5.2 Basic design. The amplifier-converter shall be mounted by normal means and tested, operating, in accordance with MIL-STD-810, Method 516.2, Procedure I, curve 516.2-2 for ground equipment.

4.8.1.5.3 Shock (transit). The drop test shall be limited to 4 feet for the equipment packaged for shipment. This test shall be performed in accordance with Method 516.2, Procedure II, of MIL-STD-810.

4.8.1.6 Dust test. The amplifier-converter shall be subjected to test in accordance with Method 510.1, Procedure I, of MIL-STD-810. Equipment shall be operated during this test.

4.8.1.7 Leakage (immersion). The amplifier-converter shall be tested in accordance with test Method 512.1, Procedure I of MIL-STD-810.

4.8.1.8 Rain. The amplifier-converter shall be tested in accordance with Method 506.1, Procedure I, MIL-STD-810, and shall be operating during last 10 minutes.

4.8.1.9 Salt fog. The amplifier-converter and all other contractor furnished items shall be subjected to test in Method 509.1, Procedure I, of MIL-STD-810. Operation of the equipment shall not be required. Failure criteria shall include corrosion of finishes and metals. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flakey, pitted, blistered or otherwise loosened finish or metal surfaces.

4.8.1.10 Fungus. The amplifier-converter and all other contractor furnished items shall be subjected to test in Method 508.1, Procedure I of MIL-STD-810. All gasket sealed or other nonhermetically sealed equipments shall be sprayed with spore suspension and exposed with the chassis removed and open from the case. Operation of the equipment shall not be required. At the conclusion of the exposure period, examination shall reveal no evidence of viable fungus or corrosion on any surfaces.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

4.8.2 Amplifier-converter. The requirements of 3.3.1.1 shall be demonstrated through a sucessful system demonstration test of radio set AN/GRC-193 and appropriate teletype equipment.

4.8.3 Control elements. Provision and operation of the control elements specified in 3.3.1.1.1 shall be demonstrated during the system test of 4.8.2.

4.8.4 Front panel connectors. Provision and operation of the front panel connectors specified in 3.3.1.1.2 shall be demonstrated during the system test of 4.8.2.

4.8.5 Fuseholders. Provision of the fuseholders specified in 3.3.1.1.3 shall be demonstrated during the system test of 4.8.2.

4.8.6 Test points - TTY. Provision of the test points specified in 3.3.1.1.4 shall be demonstrated during the system test of 4.8.2.

4.8.7 Loudspeaker. Provision of the loudspeaker specified in 3.3.1.1.5 shall be demonstrated during the system test of 4.8.2.

4.8.8 Rear mounted connectors. Provision of and proper operation of the rear mounted connectors shall be demonstrated by successful completion of the system test specified in 4.8.2.

4.8.9 Reliability. The contractor shall conduct a reliability qualification test on four first articles as selected by the Government. The test shall be conducted in accordance with Test Plan III of MIL-STD-781. Test level C of MIL-STD-781 shall be the test environment.

4.8.10 Reverse polarity. The requirements of 3.3.1.1.8 shall be tested to demonstrate no damage other than a blown fuse for reverse polarity.

4.8.11 Ground. The requirements of 3.3.1.1.9 shall be verified by observation of the applied power.

4.8.12 Test points. The requirements of 3.3.1.1.10 shall be verified by observation during performance of the maintainability demonstration test of 4.8.20.

4.8.13 Compatibility. The requirements of 3.3.1.1.11 shall be demonstrated during the system demonstration test of 4.8.2.

4.8.14 Interchangeability. The requirements of 3.3.1.1.12 shall be demonstrated by interchange of like assemblies and equipment housing.

4.8.15 Warm-up time. The requirements of 3.3.1.1.13 shall be demonstrated by observation of the time from power up to TTY Send/Receive operation as specified in 3.6.1.5. No noticable delay shall be evident.

4.8.16 Duty cycle. The requirements of 3.3.1.1.14 shall be demonstrated by observation of proper equipment performance during the burn-in test of 4.8.32.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

4.8.17 Control life. The control life requirements of 3.3.2 shall be tested on an assembled amplifier-converter.

4.8.18 Control end stops. The control end stops requirement of 3.3.3 shall be tested on an assembled amplifier-converter.

4.8.19 Covers and seals. The requirements of 3.3.4 shall be demonstrated by fully attaching and removing 25 times all covers, seals and any accessing components or cables normally used with the equipment prior to the leakage test of 4.8.1.7.

4.8.20 Maintainability. The maintainability requirements of 3.4 shall be demonstrated using contractor technicians normally assigned to like equipment.

4.8.21 Input power. The equipment shall be operated and tested with input voltages of 22.0 to 30 VDC to demonstrate compliance with 3.6.1.1.

4.8.22 Transient protection. A 10 V, 50 ms pulse, recurrent at 1 Hz, superimposed upon a 30 VDC power input shall not exceed 36 V for more than 100 microseconds per cycle measured across a 240 ohm resistor substituting for the RT-1209 load on the conditioned source.

4.8.23 Amplifier input/output. Amplifier tests for frequency response, output power, and distortion specified in 3.6.1.3.2.2 shall be made with an 8 ohm resistive dummy load substituting for the loudspeaker with the amplifier input as specified in 3.6.1.3.2.1.

4.8.24 Squelch. The requirements of 3.6.1.4, 3.6.1.4.1, 3.6.1.4.2, 3.6.1.4.3 and 3.6.1.4.4 shall tested in conjunction with an interconnected RT-1209 receiving a modulated RF signal source. A voice signal shall be simulated by square wave modulation of the 1 kilohertz (Khz) received audio at a 5 Hz rate.

4.8.25 Receive input/output. Teletype converter requirements of 3.6.1.5.1 shall be tested utilizing an interconnected RT-1209 with a suitably modulated RF signal source on the radio side and a 390-ohm resistive dummy load in the DC loop to simulate the teletypewriter load (115 ohms) and 1 mile of WD-1/TT field wire.

4.8.26 DC loop characteristics. The DC loop characteristics of 3.6.1.5.2 shall be demonstrated during, or as a part of, the teletype converter tests of 4.8.25 and 4.8.27.

4.8.27 Transmit input/output. Teletype converter requirements of 3.6.1.5.3 shall be tested utilizing an interconected RT-1209 on the radio side and a 390 ohm resistive dummy load in the DC loop to simulate the teletypewriter load (115 ohms) and 1 mile of WD-1/TT field wire.

4.8.28 Remoted teletypewriter operation. Demonstration of the requirements of 3.6.1.5.4 shall be made using a AN/TGC-14A(V) or equivalent approved substitution.

4.8.29 EMC. The system requirements of 3.8 shall be demonstrated as part of the system level demonstration required by 4.8.2. The amplifier-converter shall be subjected to test defined in MIL-STD-461, Class A3 equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

4.8.30 Dimensions. The requirements of 3.9 shall be verified as part of the mechanical inspection performed under 4.5.2.

4.8.31 Weight. The weight requirements of 3.10 shall be verified as part of the mechanical inspection performed under 4.5.2.

4.8.32 Burn-in. The burn-in test shall consist of 48 hours, with the last 24 hours of failure-free operation during which the equipment shall be operated in the TTY mode, 1:1 send/receive duty cycle for the first 12 hours and in the speaker mode, 1:1 squelch/unsquelch duty cycle for the next 12 hours; this cycle shall be repeated in the last 24 hours and shall include a complete functioning of all controls and switches. The equipment shall not be accepted if any failure occurs.

4.8.33 Workmanship screen. Prior to conducting temperature cycling test, vibration shall be performed on each equipment. The vibration may be performed at the module, drawer, or end item level. All the hardware, including cables and connectors, shall be exposed to vibration.

4.8.33.1 Vibration testing. The vibration shall be random, or subject to procuring activity approval, pseudorandum or complex wave-form vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All test items shall be hard mounted (without shock isolators) and subjected to the vibration conditions of Figure 1. The control accelerometer shall be located next to one of the mounting points of the item under test. Equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analysis of the acceleration spectral density.

4.8.33.2 Test item. The test item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during test shall be corrected and test resumed.

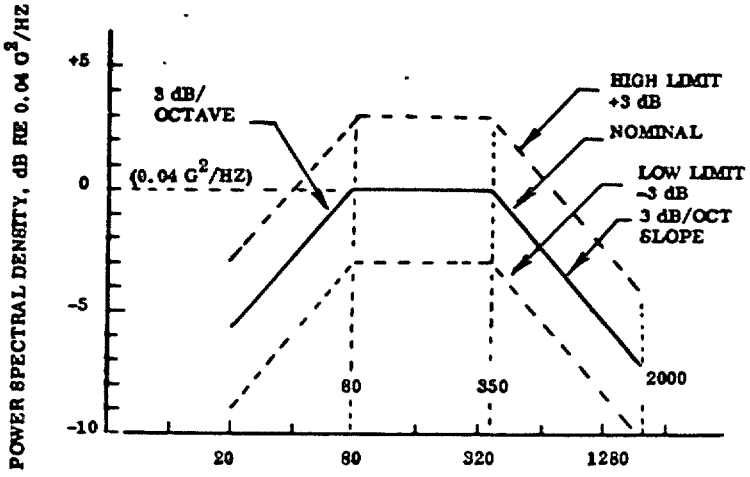


Figure 1. Random Vibration Curve

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

4.8.34 Temperature cycling. Each equipment shall be subjected to the temperature curve in Figure 2. The number of cycles required shall be determined by the electric/electronic parts count and the parts count chart below. Equipment power shall be turned on and off at the indicated times. When practical, the equipment drawers, panels and enclosures shall be opened or removed for maximum exposure to the changing temperature. The last temperature cycle shall be failure free. When performance measurements are called for, a functional operating test shall be performed. The dwell time shall be 80 percent of the time required for the largest electrical/electronic part to become temperature stabilized. Temperature stabilization shall be as defined in MIL-STD-810 for the nonoperating condition.

Electric/Electronic Parts Count	No. of Temperature Cycles
500 or less	3
501 to 1000	4
1001 to 2000	6
2001 to 3000	8
3001 and up	10

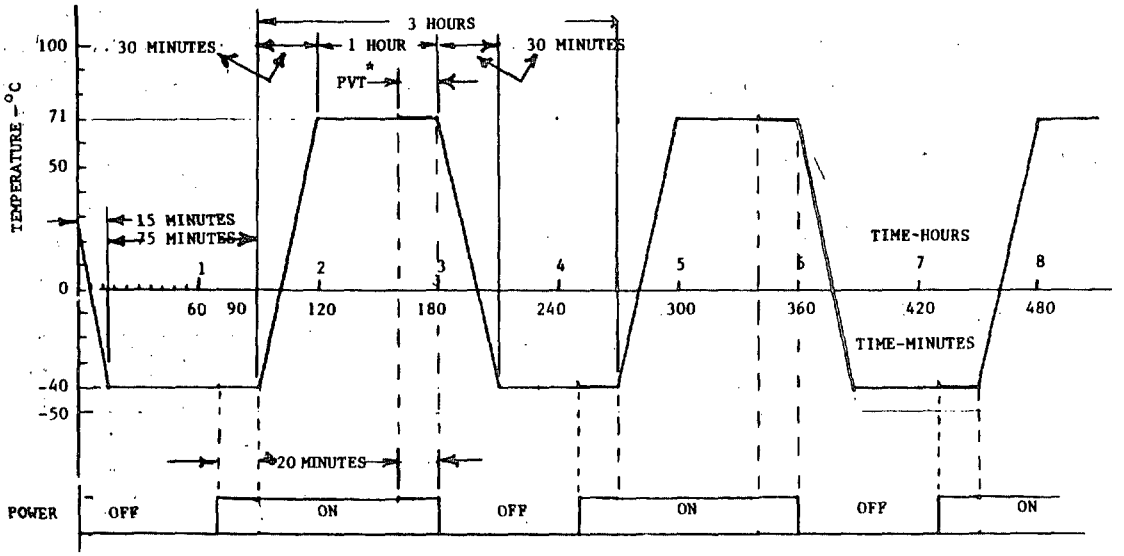


Figure 2. Temperature Cycling Curve

4.9 Performance measurements. Before, during and after the environmental tests, the following measurements shall be performed as a minimum Performance results and recorded data specifically required under the individual tests shall be provided.

Measurement	
Input power	4.8.21
Amplifier input/output	4.8.23
Squelch	4.8.24
Receive input/output	4.8.25
Transmit input/output	4.8.27

The Government reserves the right to perform any test listed at any point in the sequence of the environmental inspections. Failure to satisfy any requirement shall be a basis for rejection. A failure shall be defined as any departure from the performance required by this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation, packaging, and packing. Preservation, packaging and packing shall be the level of shipment specified in the contract or order in accordance with MIL-E-17555 (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The amplifier-converter is intended for use as part of radio set AN/GRC-193 by military forces engaged in tactical missions to establish and maintain communications with their headquarters, adjacent units, and other forces equipped with like radio sets.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) When materials, processes, and parts are other than as specified (see 3.2.1).
(c) When inspection conditions are other than as specified (see 4.3).
(d) When reinspection of conforming environmental sample units is required (see 4.7.2).
(e) Levels of preservation, packaging, and packing required (see 5.1).

5.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - MC
Project No. 5820-N811

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-29420(MC)	AMPLIFIER-CONVERTER: AM-6879/URC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:02Z
Check the source to verify that this is the current version before use.